Speckle and Multi-looking

1 Introduction

The backscattered power depends on the roughness of the surface, on the incidence angle and on the wavelength of the radar, so the same field looks different in every band. The backscattered power depends on the roughness of the surface, on the incidence angle and on the wavelength of the radar, so the same field looks different in every band. The backscattered power depends on the roughness of the surface, on the incidence angle and on the wavelength of the radar, so the same field looks different in every band. The backscattered power depends on the roughness of the surface, on the incidence angle and on the wavelength of the radar, so the same field looks different in every band. The backscattered power depends on the roughness of the surface, on the incidence angle and on the wavelength of the radar, so the same field looks different in every band.

- Average independent samples to reduce variance
- Resolution is the price of the reduction

Speckle is granular noise caused by coherent summation of many scatterers. The equivalent number of looks quantifies how much averaging took place across the image.

2 The model

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- Resolution is the price of the reduction

Multi-looking trades the resolution for radiometric accuracy over homogeneous areas of ground. The equivalent number of looks quantifies how much averaging took place across the image.

2.1 Estimating the number of looks

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1. Select a homogeneous patch
2. Compute the mean and variance
3. Divide the squared mean by the variance